

Uttkarsh Thakur

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SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C++

AI/ML: RAG Pipelines, Semantic Search, Vector Databases, FAISS, AST Parsing, NLP, Transformers, LLMs, OpenAI API

ML Libraries: LangChain, Sentence-Transformers, TensorFlow, PyTorch, Scikit-learn, OpenCV, NumPy, Pandas

Backend & Web: FastAPI, Flask, Spring Boot, Spring Data JPA, REST API Design, React.js, Node.js, HTML/CSS

Databases: PostgreSQL, MySQL, MongoDB, H2, Flyway Migrations

Developer Tools: Git & GitHub, Docker, Maven, JUnit 5, Jupyter Notebook, Google Colab, VS Code

PROJECTS

CodeNav AI | *FastAPI, React.js, LangChain, FAISS, Sentence-Transformers, OpenAI API, GitPython* 2026

- Architected a full-stack **RAG** pipeline using **LangChain** and **FAISS vector database**, enabling natural language Q&A over local codebases via **semantic search** with `all-MiniLM-L6-v2` embeddings.
- Engineered **AST-based parsing** to extract logical code units (functions/classes), preserving semantic boundaries to deliver more relevant retrieval context than naive fixed-size text chunking.
- Optimized vector index loading via **persistent disk-based caching**, cutting load times from **>20s to <100ms** — a **99.5% reduction** in latency.
- Delivered a **React.js** front-end backed by a **FastAPI** REST API for natural language codebase navigation.

BillSync | *Java 21, Spring Boot, Spring Data JPA, PostgreSQL, Flyway, React, TypeScript, Docker* 2026

- Built a full-stack expense-settlement platform with a **16-endpoint Spring Boot REST API** and a **React + TypeScript** front-end, applying layered **controller–service–repository** architecture with DTO boundaries and centralized error handling.
- Designed an exact-arithmetic split engine on **BigDecimal** guaranteeing shares sum to the total, and a greedy **dual max-heap** debt-simplification algorithm reducing any group to **at most n-1 payments** in **$O(n \log n)$** .
- Backed the system with **73 JUnit 5/Mockito tests**, **Flyway**-versioned schema migrations across **6 normalized tables**, and `@EntityGraph` query tuning that collapsed **22 SELECTs into 1**.

Motion Sensing System | *TensorFlow/Keras, VGG16, Flask, Streamlit, OpenCV, Python* 2024

- Engineered a **Human Activity Recognition (HAR)** system classifying real-time human movements using a **VGG16** model fine-tuned via **transfer learning**.
- Deployed model inference via a **Flask** REST API with a **Streamlit** dashboard for live activity prediction and visualization.

RESEARCH & TALKS

Agreement Loops in Multi-Agent Systems | *Seminar Talk, Literature Review* 2026

- Surveyed foundational and recent papers on **consensus mechanisms** in **multi-agent systems** across distributed AI research.
- Presented findings as a **seminar talk**, synthesizing convergence criteria and open challenges in agent coordination.

EDUCATION

Bennett University Aug. 2023 – May 2027

B.Tech in Computer Science and Engineering | **CGPA: 8.17/10** Greater Noida, India

- Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Deep Learning, Neural Networks, System Design

DAV Public School Graduated 2022

Senior Secondary Education (Class XII) Ranchi, India

CERTIFICATIONS

Build Basic Generative Adversarial Networks (GANs) — DeepLearning.AI via Coursera Apr. 2026

Neural Networks and Deep Learning — DeepLearning.AI via Coursera Apr. 2025

Machine Learning: Clustering & Retrieval — University of Washington via Coursera Nov. 2024

ACHIEVEMENTS

Ranked **101st nationally** at **Smart India Hackathon 2024**

CodeChef: 2-star rated | **LeetCode:** Contest rating **1500+**

Competed in **Hackaccino Hackathon**, Bennett University (2024)